Angela Qian

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EDUCATION

Purdue University | *B.S. in Computer Science and Mathematics* **Aug 2023 – May 2027**
Concentrations in Machine Intelligence and Algorithmic Foundations

EXPERIENCE

Undergraduate Research Assistant | *Advisor: Joseph Campbell* **Feb 2025 – Present**
- Currently working on a project developing hierarchical world models in model-based reinforcement learning
- Worked on a team to design an LLM-based agent with Theory of Mind to play the social deduction game Avalon
- Conducted participant studies with 45+ volunteers and applied statistical analysis to evaluate model performance
- Developed and evaluated methods for hallucination detection with 95% agreement with human annotations

Software Development Engineer Intern | *Amazon Lab126* **May 2026 – Aug 2026**
- Built an agentic ticket triage system for the Alexa Smarthome Cameras Streaming team that autonomously analyzes diagnostic logs, publishes root cause analyses, and drafts fixes as pull requests
- Developed a notification system for triage updates, enabling interactive investigation and remote PR drafting
- Added full-stack frame-rate/bitrate time series graphs to a session diagnostics dashboard, aggregating per-interval throughput end-to-end from a Java model and DynamoDB store through a worker to a React frontend

Software Engineering Intern | *Cepton Technologies Inc.* **May 2025 – Aug 2025**
- Performed real-time foreground-background segmentation masking from depth and reflectivity maps by engineering a computer vision model based on U-Net segmentation, which has been integrated into the production system
- Built in Rust a real-time object tracking pipeline using LiDAR-generated point cloud data on a per-frame basis
- Developed a 2D rectangle selection tool for a 3D point cloud viewer, enabling users to select points from any camera perspective and display their coordinates dynamically by transforming 3D point data using projection matrices

Teaching Assistant | *Purdue University Department of Computer Science* **Jan 2025 – Present**
- CS 381: Analysis of Algorithms, CS 211/311: Competitive Programming, CS 182: Discrete Mathematics
- Led weekly recitation sessions, teaching problem-solving techniques and reinforcing theory taught in class
- Held weekly office hours to clarify course material and guide students through homework challenges

Undergraduate Researcher | *Purdue University Vertically Integrated Projects* **Aug 2024 – Dec 2024**
- Detected image blur with custom program using convolution, the Laplacian filter, and the Sobel filter
- Generated clear, focused photos from blurred images with algorithms based on Fourier Transforms
- Presented in Purdue's 2024 Fall Undergraduate Research Exposition

PUBLICATIONS

Bayesian Social Deduction with Graph-Informed Language Models
S. Rahimirad, G. Gergerli, L. Romero, **A. Qian**, M. L. Olson, S. Stepputtis, J. Campbell. *ACL 2026*

PROJECTS

2048 AI Player | *PyTorch, TorchRL, Jupyter Notebook, JavaScript* **June 2025 – Present**
- Developed a neural network to play 2048 using imitation learning trained on a custom dataset of my own gameplay
- Authored and published a research-paper-styled writeup in SIGBOVIK-inspired journal, SIGHORSE
- Currently extending the project by implementing a Deep Q-Network for reinforcement learning to fine-tune strategic decision-making through self-play

ScribbleScore Handwriting Improvement | *Python, OpenCV, Numpy, React, Flask* **Feb 2025**
- Worked in a group of 4 over the course of a 36-hour hackathon (Boilermake XII)
- Developed a web app that scans handwriting from images, evaluates it from 0–100, and provides feedback
- Implemented Sobel edge detection, line of best fit analysis to assess writing straightness, extracting text from images, and computing score

SKILLS

Languages: English (Native), Mandarin Chinese (Native)
Programming: Python, Rust, C/C++, JavaScript, Typescript, x86_64 Assembly, SQL, MongoDB, HTML, CSS, Java
Developer Tools: Git, GitHub, Jupyter Notebook, Figma, Linux, Overleaf, React, Node, Bash, LaTeX, VIM, FFmpeg, MySQL, MongoDB, Neo4J, Next.js, TailwindCSS, VS Code, Eclipse, Inkscape
Libraries: Matplotlib, OpenCV, PyTorch, TorchVision, Pandas, NumPy